

Retail Customer Behavior & Revenue Analytics

End-to-End Retail Data Analysis Pipeline

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Tools Deployed:

MySQL, Microsoft Power BI, Power Query

Domain:

Retail & Consumer Goods

Project Duration:

2 Weeks

Project Overview:

This project focuses on analyzing customer transaction data from multiple shopping malls to understand purchasing behavior, revenue patterns, and key business drivers. The analysis involves data validation using SQL, data transformation through MySQL and Power Query, and visualization using Power BI. The objective is to identify high-value customer segments, top-performing product categories, and payment preferences. The project ultimately delivers actionable insights to support data-driven business decisions and improve overall performance.

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1. Executive Summary

This project presents a comprehensive analysis of customer transaction data aimed at understanding purchasing behavior, identifying revenue drivers, and uncovering actionable business insights. The dataset consists of approximately 99,000 transactions and nearly 299,000 products sold, generating a total revenue of around 68.55 million. The analysis was conducted using a structured approach involving SQL-based data validation, data transformation, exploratory analysis, and interactive visualization.

A key finding of the analysis is that female customers represent the dominant customer segment, contributing significantly higher transaction volume and revenue compared to male customers. Age-based analysis reveals that the 46-60 age group is the most valuable segment, generating the highest revenue and product purchases. Category-level insights show that Clothing is the top-performing category, followed by Shoes and Technology, indicating strong consumer demand in these segments.

In terms of payment behavior, cash remains the most preferred method, accounting for the largest share of transactions and revenue, while credit and debit cards follow. The analysis also highlights stable revenue trends over time, suggesting consistent customer demand without major seasonal fluctuations.

From a business perspective, the findings emphasize the importance of targeted marketing towards high-value segments, strategic inventory planning for top categories, and promoting digital payment adoption. Overall, this project demonstrates how structured data analysis can transform raw transactional data into meaningful insights, enabling businesses to make informed decisions and improve operational efficiency.

2. Business Problem / Objective

The business aims to understand customer purchasing behavior and identify key drivers of revenue to improve decision-making in retail operations. The focus is on analyzing transaction data to uncover patterns related to customer segments, product performance, and payment preferences.

Key Analytical Questions:

- Which customer segments contribute the most to revenue?
- Which product categories generate the highest sales and volume?
- How does payment behavior vary across customers and transactions?
- Are there identifiable patterns across age groups and gender?

Scope of Analysis:

- Analyze transaction-level data across multiple malls
- Evaluate customer demographics (age, gender)
- Assess category-wise performance (revenue and volume)
- Study payment method distribution and behavior
- Identify actionable insights for business improvement

3. Dataset Description

The dataset consists of retail transaction records collected from multiple shopping malls between 2021 and 2023. It provides detailed information on customer purchases and supports analysis of sales patterns and customer behavior.

Dataset Overview:

- Total Transactions: ~99,000
- Total Products Sold: ~299,000
- Total Revenue: ~68.55 Million

Columns Description:

- **invoice_no**: Unique identifier for each transaction
- **customer_id**: Unique identifier for each customer
- **gender**: Gender of the customer (Male/Female)
- **age**: Age of the customer
- **category**: Product category (e.g., Clothing, Shoes, Technology)
- **quantity**: Number of units purchased
- **revenue**: Total transaction value (derived from price column)
- **payment_method**: Mode of payment (Cash, Credit Card, Debit Card)
- **invoice_date**: Date of transaction
- **shopping_mall**: Location where the purchase was made

Time Range:

- The dataset covers transactions from **2021 to 2023**, enabling time-based analysis of revenue and customer trends.

4. Data Validation & Quality Checks

A structured data validation process was performed using SQL to ensure data reliability before analysis. All checks were executed directly on the database using MySQL queries.

Column Audit:

- Verified dataset structure using DESCRIBE and sample queries
- Ensured all required columns are present and consistently populated

NULL Checks:

- Performed column-wise NULL and empty value checks
- Validated critical fields such as gender, age, category, quantity, price, and payment_method
- Used aggregated SQL queries to identify missing values across all columns

Duplicate Checks:

- Checked for duplicate invoice numbers using GROUP BY and HAVING COUNT(*) > 1
- Ensured transaction uniqueness to avoid inflated counts in analysis

Data Type Validation:

- Verified appropriate data types for each column (numeric, categorical, date)
- Ensured invoice_date is usable for time-based analysis
- Confirmed quantity and price fields are numeric for calculations

Anomaly Detection:

- **Age:**
 - Identified invalid values (age ≤ 0 or > 100)
 - Checked for NULL values
 - Ensured demographic data is realistic
- **Quantity:**
 - Detected zero and negative values
 - Evaluated min, max, and average to identify abnormal transactions
 - Ensured values represent actual purchases
- **Price:**
 - Checked for NULL, zero, and negative values
 - Analyzed distribution using min, max, and average
 - Derived unit price (price / quantity) to validate consistency

Key Validation Insight:

- Derived unit price was consistent within categories, indicating that the price column represents total transaction value rather than unit price

Overall, SQL-based validation ensured that the dataset is structurally correct, consistent, and ready for transformation and analysis.

5. Data Issues Identified

During validation, two critical data issues were identified that directly impact analysis accuracy:

5.1 Price Issue (Critical)

- The price column does not represent unit price
- Instead, it represents the total transaction value (revenue)

Evidence:

- Derived unit price (price / quantity) remained consistent across records
- Confirms that price is already aggregated at transaction level

Why this is a problem:

- Misinterpreting price as unit price would lead to:
 - Incorrect revenue calculations
 - Double-counting during aggregation
 - Misleading category performance analysis

Resolution:

- Column was correctly reinterpreted and renamed as revenue for analysis consistency

5.2 Age Issue

- Age values were available as raw numerical data
- No predefined grouping or segmentation

Why this is a problem:

- Raw age values do not provide meaningful business insights
- Difficult to:
 - Identify target customer segments
 - Compare behavior across demographics
 - Perform aggregated analysis

Impact:

- Without segmentation, patterns like “which age group drives revenue” cannot be identified clearly

Resolution:

- Age was transformed into structured groups (e.g., 18-25, 26-35, 36-45, 46-60, 60+)

5.3 Summary of Impact

These issues are critical because they directly affect analysis accuracy. If not handled properly:

- Revenue calculations would be incorrect
- Customer segmentation would be ineffective
- Business decisions would be based on misleading insights

Identifying and resolving these issues ensured that the dataset is analytically reliable and suitable for meaningful business analysis.

6. Data Transformation

Data transformation was performed using both MySQL and Power Query to prepare the dataset for accurate and structured analysis.

6.1 MySQL Transformation

Transformations were applied using a SELECT-only approach to avoid modifying the original dataset.

- **Price → Revenue (Alias):**
 - The price column was aliased as revenue since it represents total transaction value
 - Ensures correct interpretation during aggregation and reporting
- **Age Category Creation:**
 - Created a new column age_category using a CASE statement
 - Segmented customers into groups:
 - 18-25, 26-35, 36-45, 46-60, 60+
 - Enabled structured demographic analysis
- **Final Transformed Dataset:**
 - Combined all required columns along with derived fields
 - Exported as a clean analytical dataset for further processing

6.2 Power Query Transformation

Additional transformations were performed in Power BI (Power Query) for reporting and visualization.

- **Age Group Binning:**
 - Recreated structured age groups for consistency in visuals
- **Time-Based Feature Extraction:**
 - Extracted **Year** and **Month** from `invoice_date`
 - Enabled trend and time-series analysis
- **Categorical Data Cleaning:**
 - Standardized text fields (e.g., `category`, `payment_method`)
 - Removed inconsistencies and ensured uniform naming

These transformations ensured that the dataset is clean, structured, and optimized for analytical use in Power BI.

7. Feature Engineering

Feature engineering was performed to enhance the dataset and enable deeper analytical insights.

7.1 Derived Features

- **Revenue:**
 - Derived from the original price column
 - Represents total transaction value
 - Used as the primary metric for business analysis
- **Age Group (age_category / age_group):**
 - Created by grouping raw age values into meaningful segments
 - Enables demographic comparison and customer segmentation
- **Time-Based Features:**
 - Year and Month extracted from transaction date
 - Supports trend analysis and performance tracking over time

7.2 Purpose of Feature Engineering

These features were created to:

- Improve interpretability of raw data
- Enable meaningful aggregation and comparison
- Support business-focused analysis (e.g., segment-wise revenue)
- Enhance visualization and reporting capabilities

Overall, feature engineering transformed raw transactional data into a structured format suitable for extracting actionable insights.

8. Exploratory Data Analysis (EDA)

Initial analysis was performed to understand overall data distribution and identify high-level patterns before detailed segmentation.

Overall Metrics:

- Total Revenue: ~68.55M
- Total Transactions: ~99K
- Total Quantity Sold: ~299K
- Average Order Value: ~689

Key Observations:

- Revenue distribution is relatively stable across the dataset
- No extreme spikes or irregularities in transaction patterns
- Product sales are consistently distributed across categories
- Majority of transactions involve moderate quantity levels
- Dataset shows a balanced mix of customer segments and purchase behavior

These observations indicate that the dataset is well-structured and suitable for deeper analysis without significant skew or instability.

9. Visualization Analysis

Detailed analysis was performed using Power BI dashboards to extract insights across key business dimensions.

9.1 Gender Analysis

- Female customers account for ~60% of total transactions
- Male customers contribute ~40%
- Female customers purchased ~179K products vs ~120K by males
- Revenue contribution:
 - Female: ~41M
 - Male: ~28M

Insight:

Female customers are the dominant segment across transactions, volume, and revenue, making them the primary revenue drivers.

9.2 Age Group Analysis

- Highest transactions from age group 46-60 (~28K)
- Followed by 36-45 and 26-35 groups
- Product volume:
 - 46-60: ~85K (highest)
- Revenue contribution:
 - 46-60: ~19.6M (highest)

Insight:

Age group 46-60 is the most valuable customer segment in terms of engagement, volume, and revenue.

9.3 Category Analysis

- Top revenue-generating categories:
 - Clothing (~31M)
 - Shoes (~18M)
 - Technology (~15.8M)
- Other categories (Books, Souvenir, Food & Beverage) contribute significantly less
- Clothing shows strong demand across all demographics
- Female customers contribute more revenue across most categories

Insight:

A small set of categories (Clothing, Shoes, Technology) drives the majority of business revenue.

9.4 Payment Analysis

- Transaction distribution:
 - Cash: ~44.69%
 - Credit Card: ~35%
 - Debit Card: ~20%
- Revenue contribution follows similar pattern:
 - Cash generates highest revenue
 - Credit card is second
 - Debit card is lowest
- Female customers dominate across all payment methods
- Age group 46-60 shows higher usage across all payment types

Insight:

Cash remains the most preferred payment method despite digital alternatives.

9.5 Time Trend Analysis

- Revenue trend from 2021 to 2023 remains stable
- Monthly performance shows consistent sales with no extreme fluctuations
- Indicates steady demand across the period

Insight:

Business performance is stable over time, with no strong seasonality or volatility.

10. Key Insights

- Female customers contribute ~60% of total transactions and generate ~41M revenue, making them the primary revenue-driving segment.
- Male customers contribute ~40% of transactions and ~28M revenue, indicating a significantly lower but still consistent contribution.
- The age group 46-60 records the highest engagement with ~28K transactions, ~85K products sold, and ~19.6M revenue.
- Age groups 36-45 and 26-35 show similar performance, each contributing over ~13M revenue, forming strong secondary segments.
- Younger customers (18-25) generate the lowest revenue (~10.4M), indicating lower purchasing power or engagement.
- The Clothing category is the top performer, generating ~31M revenue, significantly higher than all other categories.
- Shoes (~18M) and Technology (~15.8M) are the next highest revenue contributors, forming the core revenue-driving categories.
- Low-performing categories such as Books, Souvenir, and Food & Beverage contribute less than ~1M each, indicating limited business impact.
- Female customers consistently outperform males across all major categories, especially in Clothing, Shoes, and Technology.
- Cash accounts for ~44.69% of all transactions, making it the most preferred payment method despite digital alternatives.
- Credit cards contribute ~35% of transactions and generate strong revenue, indicating growing digital adoption.
- Revenue trends remain stable from 2021 to 2023 with no major seasonal spikes, indicating consistent customer demand.

11. Business Recommendations

- **Targeted Marketing:**
Focus campaigns on female customers and the 46-60 age group, as they generate the highest revenue. Use personalized offers, email campaigns, and in-store promotions to maximize conversion.
- **Category-Focused Promotions:**
Prioritize Clothing, Shoes, and Technology in advertisements and seasonal campaigns, as these categories drive the majority of revenue.
- **Inventory Optimization:**
Maintain higher stock levels for top-performing categories (Clothing, Shoes, Technology) to avoid stockouts. Reduce overstock in low-performing categories like Books and Souvenir.
- **Segment-Based Offers:**
Design age-specific offers (e.g., premium products for 46-60, discounts for younger segments) to improve engagement across all customer groups.
- **Payment Strategy Improvement:**
Encourage digital payments (Credit/Debit cards) through cashback offers, discounts, or reward points to gradually reduce dependency on cash.
- **Loyalty Programs:**
Introduce loyalty programs targeting high-value customers (especially 46-60 age group) to increase repeat purchases and long-term retention.
- **Cross-Selling Opportunities:**
Bundle popular categories (e.g., Clothing + Shoes) to increase average order value and improve customer purchase experience.

12. Business Impact

- **Revenue Growth:**
Focusing on high-performing segments and categories can significantly increase revenue by improving conversion rates and average order value.
- **Improved Customer Engagement:**
Personalized marketing and loyalty programs enhance customer experience, leading to higher retention and repeat purchases.
- **Better Inventory Management:**
Aligning inventory with demand reduces stockouts and excess inventory, improving operational efficiency and reducing costs.
- **Increased Sales Efficiency:**
Category-focused strategies and cross-selling can boost sales without increasing customer acquisition costs.
- **Digital Payment Adoption:**
Encouraging digital payments improves transaction efficiency, reduces cash handling risks, and supports modern retail practices.

Overall, these actions enable the business to move towards a more data-driven approach, improving both short-term performance and long-term growth.

13. Limitations

- **No Profitability Data:**
The dataset includes revenue but does not contain cost or profit information, limiting the ability to perform margin or profitability analysis.
- **No Customer-Level Tracking:**
Lack of detailed customer history prevents analysis of repeat purchases, customer lifetime value, and retention behavior.
- **Limited Scope of Dataset:**
Data is restricted to selected malls and a specific time period (2021-2023), which may not fully represent broader market trends.
- **No External Factors:**
The dataset does not include external variables such as promotions, seasonality drivers, or economic conditions that may influence purchasing behavior.

14. Future Scope

- **Customer Segmentation using Machine Learning:**
Apply clustering techniques (e.g., K-Means) to identify distinct customer groups based on purchasing behavior.
- **Predictive Analysis:**
Build forecasting models to predict future sales, demand trends, and customer behavior.
- **Recommendation Systems:**
Implement product recommendation engines to enhance cross-selling and personalized shopping experiences.
- **Profitability Analysis:**
Incorporate cost data to analyze profit margins and optimize pricing strategies.
- **Advanced Time-Series Analysis:**
Perform deeper seasonal and trend analysis for better planning and forecasting.

15. Conclusion

This project successfully analyzed large-scale retail transaction data to uncover meaningful insights into customer behavior, revenue drivers, and business performance. The analysis identified that female customers and the 46-60 age group are the most valuable segments, while Clothing, Shoes, and Technology are the primary revenue-generating categories. Payment analysis highlighted the continued dominance of cash, alongside growing digital adoption.

By applying structured data validation, transformation, and visualization techniques, the project demonstrates how raw data can be converted into actionable business insights. The recommendations provided can support targeted marketing, efficient inventory planning, and improved customer engagement.

Overall, this project highlights the importance of data-driven decision-making in retail and provides a strong foundation for future analytical and strategic initiatives.